Length of circular arc

1

a 11.5 units

b 19.6 units

2

a 10.47 cm

b 33.25 mm

c 9.42 cm

d 6.00 cm

3

a $\frac{5\pi}{6}$ radians

b $\frac{25\pi}{2}$ cm

4 $\left(20 + \frac{25\pi}{3}\right)$ cm

5

a 36.8 cm

b 16.8 cm

6 $\theta = 1.95$ radians

7

a $\theta = 0.9$ radians

b $\theta = 1.1$ radians

8

a $\theta = 1.67$ radians

b $\theta = 96^\circ$

9 70π cm

Area of a sector

10

a 463.10 m^2

b 313.01 m^2

11 39π units^2

12

a 50 cm^2

b 729.87 m^2

c 612.42 m^2

13

a $\frac{2\pi}{3}$

b 108π cm^2

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14

a $\theta = 2$ radians

b $\frac{9}{4}\pi^2 \text{ cm}^2$

15

a $6\pi \text{ cm}^2$

b $6\pi \text{ cm}^2$

c $6\pi \text{ cm}^2$

16

a $\angle JOK = 46^\circ$

b 90 cm^2

17 81 cm^2

18 317.9 m^2

19

a $48\pi \text{ cm}^2$

b $336\pi \text{ cm}^2$

20

a $10\pi \text{ cm}^2$

b $(5\pi + 8) \text{ cm}$

21 $\theta = 1.7$ radians

22

a $\theta = 0.8$ radians

b $l = 16 \text{ m}$

23 $l = 11 \text{ cm}$

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Triangles in sectors

24

a $\angle AOC = 0.76$ radians

b Arc $AB = 44.08$ cm

25

a $AB = 12$ cm

b $\left(3\sqrt{2}\pi - 12\right)$ cm

26

a $AB = 7.2$ cm

b $x = 7.0$ cm

c 28.8 cm^2

27

a 49 cm^2

b $\left(\frac{49\pi}{3} - 49\right)$ cm^2

28 $\left(18\pi - 36\sqrt{2}\right)$ cm^2

29

a

i 61.99 cm^2

ii 383.33 cm^2

b

i 126.90 cm^2

ii 18.90 cm^2

30

a $\angle ROP = 1.02$ radians

b 42.43 cm

31 $\left(100\sqrt{3} - 50\pi\right)$ cm^2

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